



EX-EMPLOYEE VERIFICATION PORTAL

*A secure, automated platform designed to streamline employment verification
workflows between organizations and HR teams.*

Built for the CodeMate Community

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01. PROJECT AT A GLANCE

The Ex-Employee Verification Portal is a full-stack Next.js application designed to securely verify employment details of former employees. It streamlines communication between third-party verification agencies and internal HR departments.

The platform supports two primary user groups:

- **Verifiers:** External companies or background verification agencies that submit employee verification requests.
- **Admin / HR Users:** Internal HR or administrative users who review verification activity, handle appeals, inspect access logs and export verification data.

The application combines authentication, OTP-based login, employee verification, appeals handling, PDF report generation, email notifications, access auditing, and supporting-document uploads into one centralized platform.

Core Capabilities

- Verifier Registration & Login (OTP-Based 2FA)
- Field-Level Comparison & Match Scoring
- Appeal Submission & Document Upload
- Admin Dashboard & Appeal Management
- Audit Logging & Access Logs
- Excel-Friendly Export & PDF Reports
- Email Notifications & Statistics
- Role-Based Access Control

02. PROBLEM, USERS & PRODUCT VALUE

The Problem

Employee verification is often handled through manual communication between companies, verification agencies, and HR teams. This can make it difficult to validate employee information, track verification requests, manage mismatches, handle appeals, and maintain a reliable audit history.

Target Users

User Type	Use Case
Verifiers	Register, submit employee details, view match results, and raise appeals with supporting documents.
Admin / HR Users	Manage operations, review appeals, inspect access logs, and export verification data.

Product Value

- Centralized verification workflow and OTP-based authentication.
- Automated field-by-field comparison and match scoring.
- Structured appeal management and supporting document handling.
- Comprehensive PDF reporting and email notification system.
- Secure audit logging and role-based access.

03. CUSTOMER PERSONA

Primary Persona: HR / Background Verification Manager

Goal: Review and respond to employee verification requests efficiently while maintaining a reliable record of every verification and appeal.

Needs

- Quickly review verification requests and validate employee information.
- Identify mismatched fields and track results.
- Review supporting documents and handle appeals effectively.

Pain Points

- Manual, fragmented verification processes.
- Scattered employee information and unclear appeal statuses.
- Manual document exchange and lack of audit trails.

04. ARCHITECTURE & TECHNOLOGY

The portal uses a modern full-stack architecture with dedicated service layers and persistence.

Layer	Technology Stack
Frontend	Next.js 16, React 19, Tailwind CSS 4, DaisyUI, Framer Motion
Backend	Next.js API Routes, Node.js, JWT Auth, Joi Validation
Database	MongoDB with Mongoose ORM
Reporting	jsPDF, jsPDF-AutoTable, html2canvas, Recharts
Security	bcryptjs, Helmet, CORS, Rate Limiting, RBAC

Architecture Flow

User → Next.js / React → API Routes → Services → Mongoose → MongoDB

05. CORE VERIFICATION WORKFLOW

The main workflow connects verifier authentication, employee validation, verification results, appeals, and reporting.

Verifier Workflow

Registration → Login → OTP Verification → Verification Wizard → Employee Validation → Data Comparison → Verification Result → PDF / Appeal

Admin Workflow

Admin Login → Dashboard → Appeals / Logs → Review Appeal → HR Response → Export / Reporting

06. FEATURE MAP

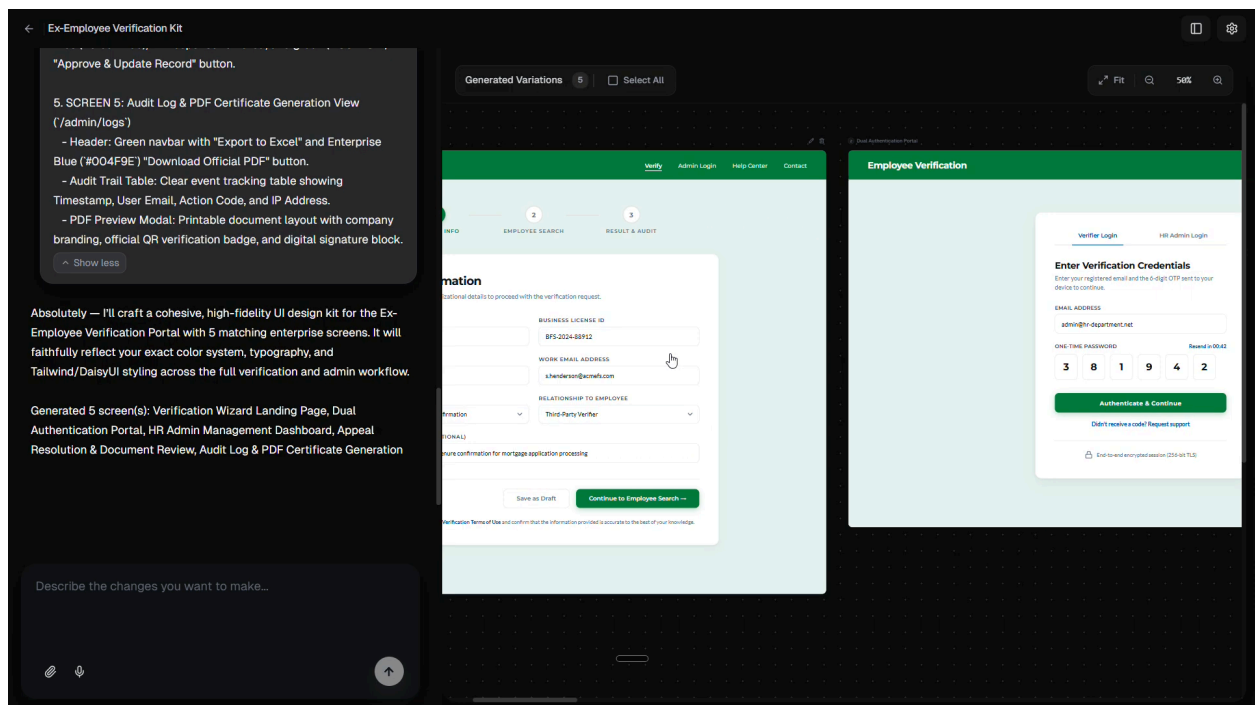
Feature	Description
Verification Wizard	Multi-step interface for submitting employee information.
Comparison Engine	Automated matching for ID, Name, Designation, and Exit Reason.
Match Scoring	Classification system for Matched, Partial Match, or Mismatch results.
Appeals System	Verifier interface for disputing results with document uploads.
Admin Dashboard	KPI cards for total verifications, pending appeals, and system health.
Access Logs	Detailed audit trails including timestamps, IP addresses, and user-agents.
Reporting	Official PDF report generation and Excel-friendly CSV exports.

07. BUILD: DESIGN MODE

Objective: Create a cohesive, high-fidelity UI design system for the Ex-Employee Verification Portal with a professional enterprise verification experience.

Design Direction

- **Primary Brand:** Forest Green (#007A3D) for headers, navigation, and primary CTAs.
- **Secondary Accent:** Enterprise Blue (#004F9E) for admin logins and information tabs.
- **Aesthetic:** Soft Mint backgrounds, pure white cards, and Montserrat/Lato typography.
- **UI Elements:** Rounded cards, DaisyUI components, and clear step indicators.

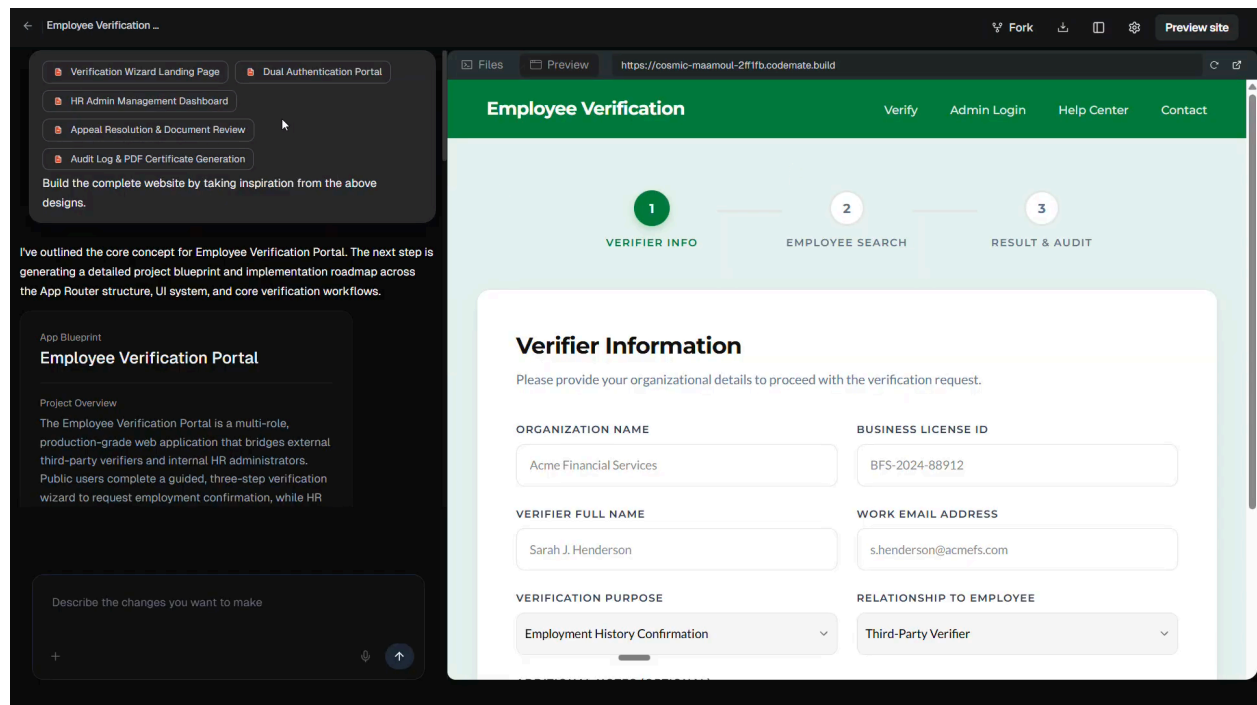


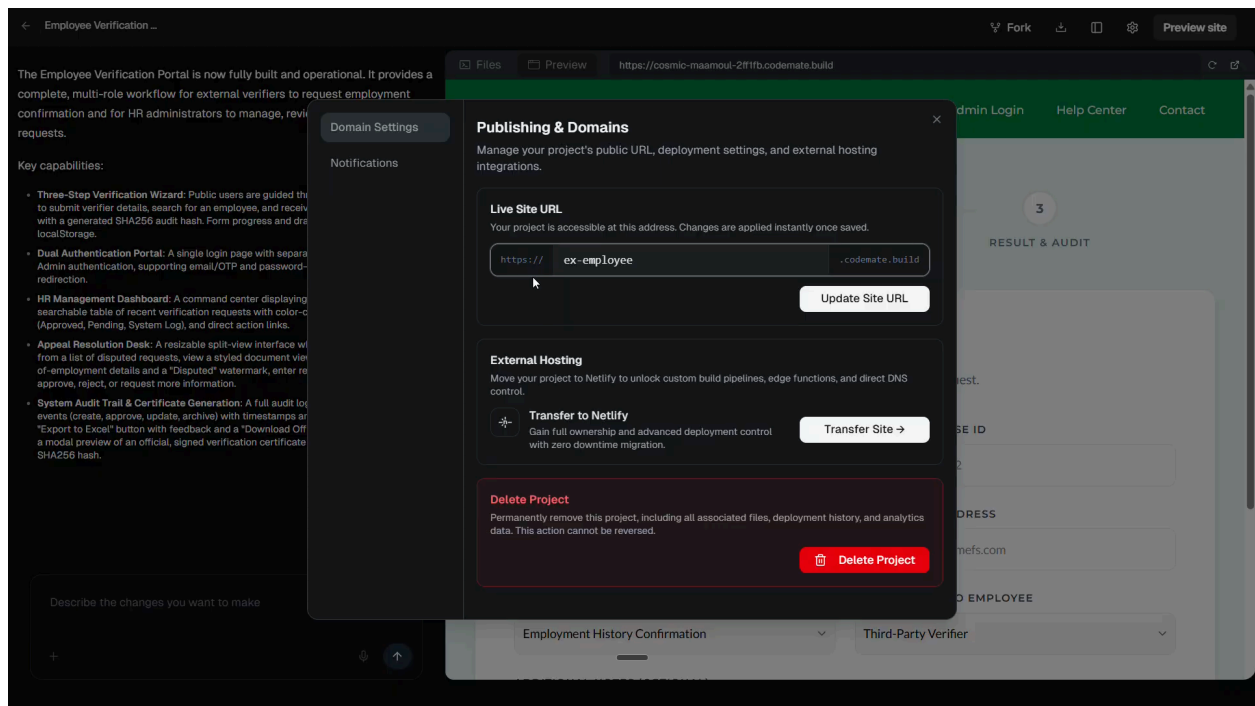
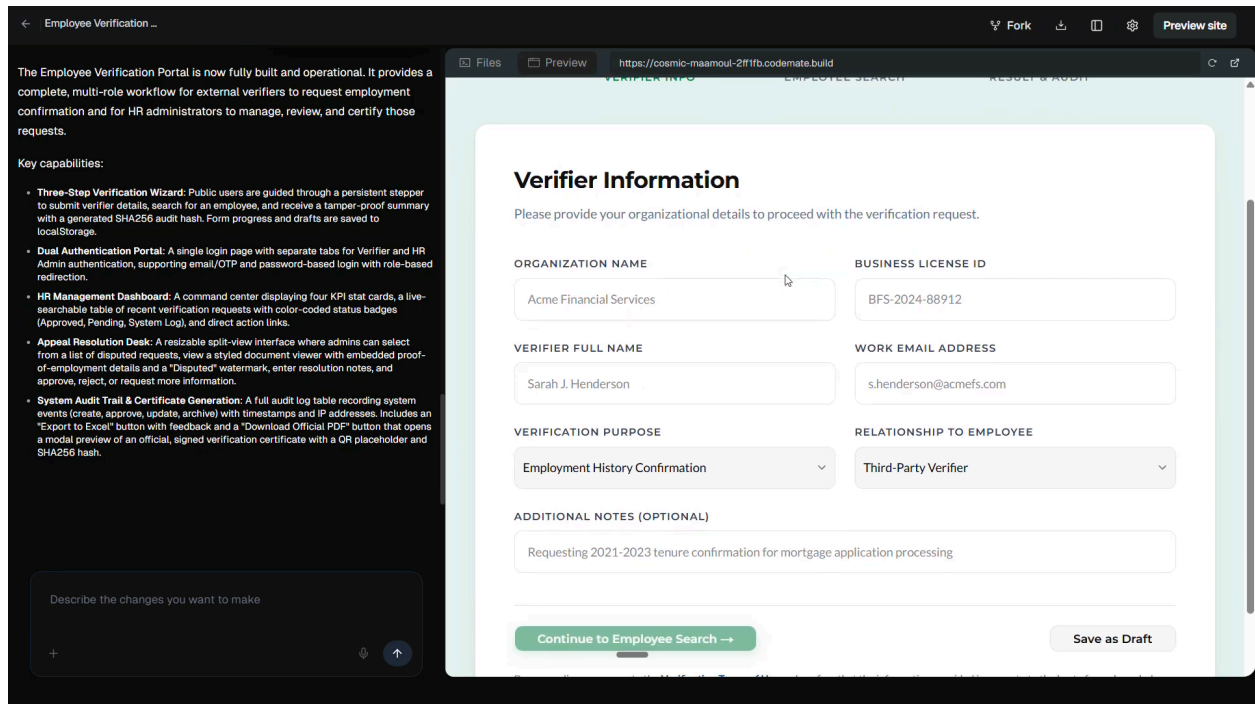
08. BUILD: PROTOTYPE MODE

Objective: Convert the approved design into a functional full-stack application while maintaining the established enterprise visual system.

Implementation Scope

- Authentication: Verifier/Admin login, OTP verification, and JWT session handling.
- Core Logic: Verification wizard, result comparison service, and appeal management.
- Admin Tools: Dashboard cards, audit logs, and data export.
- Shared UI: Responsive layouts, status badges, and reusable form components.





09. BUILD PROMPT HIGHLIGHTS

- **Prompt 1 : Verification Portal Design**

Act as a Principal UI/UX Designer. Reverse-engineer and generate a cohesive, high-fidelity UI design kit with 5 key screens for the "Ex-Employee Verification Portal", matching the exact color scheme and branding of the existing web application. Use Forest Green (#007A3D) as the primary brand color, Enterprise Blue (#004F9E) as the secondary accent, white and soft mint backgrounds, Montserrat for headings and Lato for body text. Use Tailwind CSS and DaisyUI with rounded enterprise cards, clear step indicators and structured tables. This is based directly on the actual Design Mode prompt sheet.

- **Prompt 2 : Verification Wizard**

Create the Ex-Employee Verification Portal verification experience with a professional multi-step wizard. Include verifier information, employee search and verification result stages. Use clear step indicators, structured employee input fields, green focus states, validation feedback and a clear final result state. The interface should allow authenticated verifiers to submit employee details and understand the verification status and available next actions.

- **Prompt 3 : Admin Management**

Create the HR Admin Management experience for the Ex-Employee Verification Portal. Include an admin dashboard with KPI cards for total verifications, pending appeals, approved cases and system logs. Add structured tables for verification activity, appeals and access logs, with clear status badges, filters and enterprise-style data presentation. Include actions for reviewing appeals, inspecting documents and exporting verification information. The actual Design Mode specification includes these dashboard KPI cards, tables, status badges and admin workflows.

- **Prompt 4 : Appeals, Audit & Reporting**

Extend the Ex-Employee Verification Portal with an appeal-resolution and reporting experience. Create a split-panel appeal review interface where HR users can inspect disputed verification requests, view uploaded supporting documents, review verifier comments and submit an approval or rejection response. Also provide an audit-log interface showing timestamps, user identity, action codes and IP information, together with an official PDF verification preview and export actions.

10. CORA: PLANNING & IMPLEMENTATION

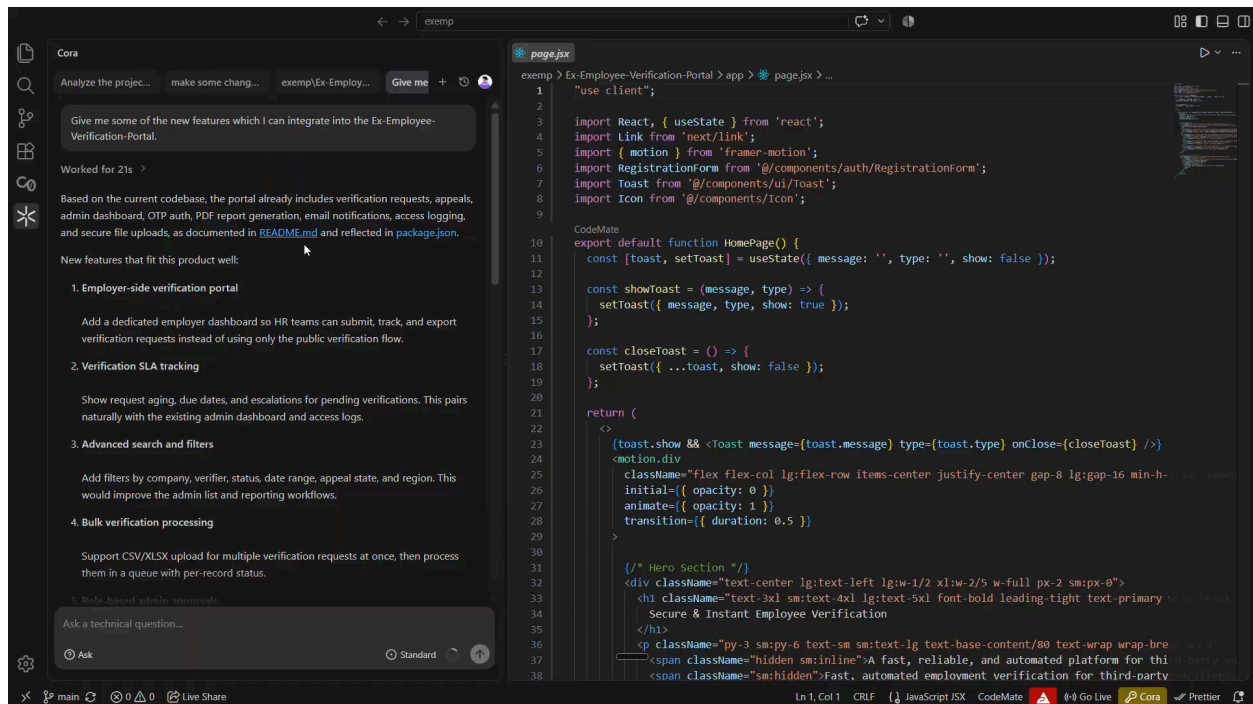
CORA was utilized for backend API development, MongoDB integration, and complex logic orchestration.

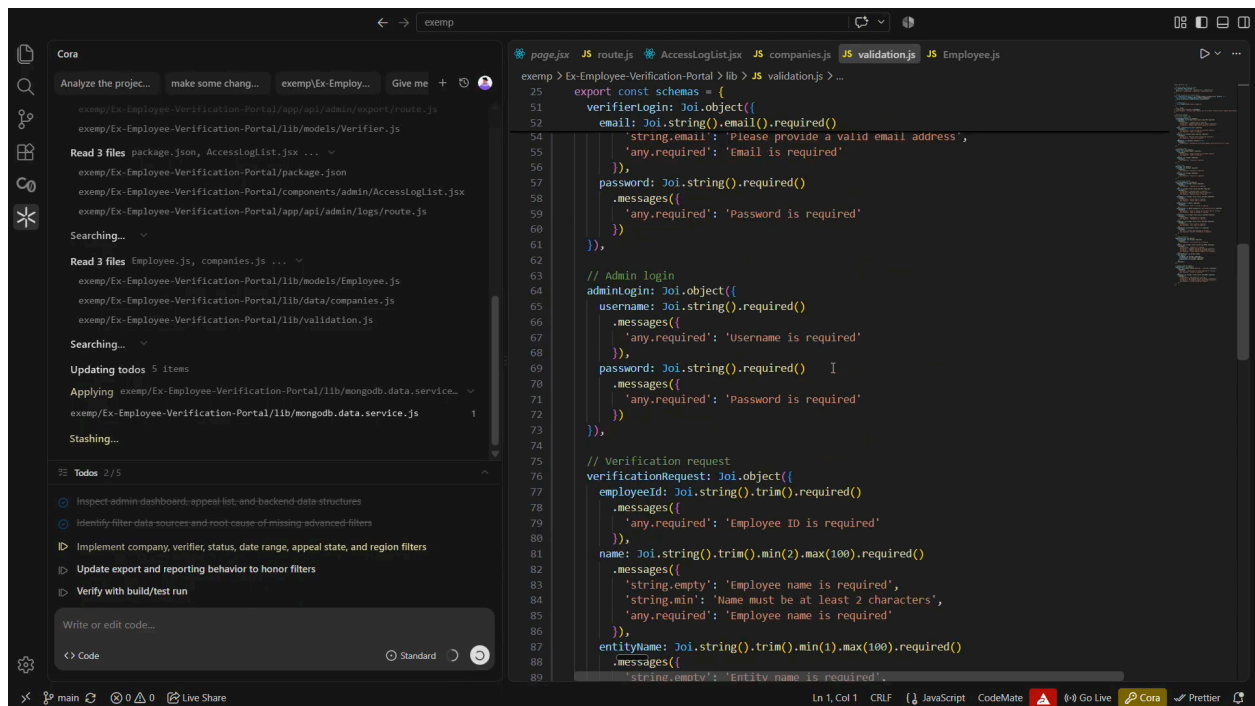
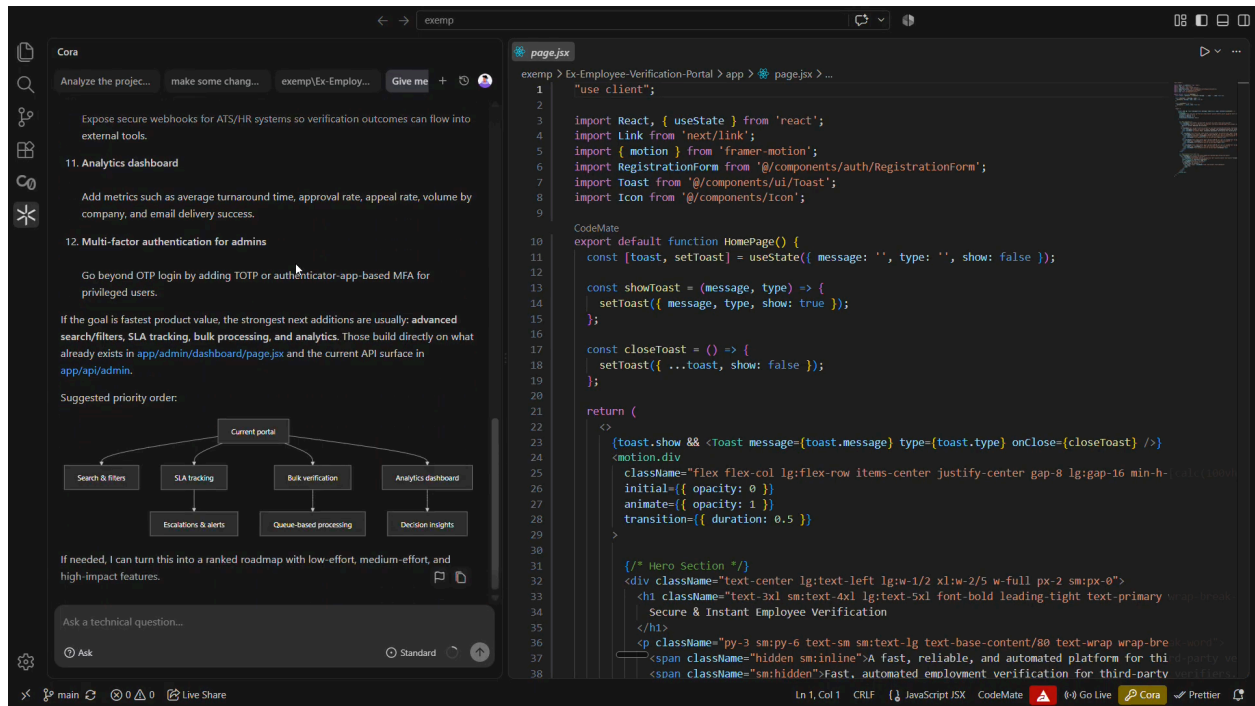
Workflow

Requirement → CORA Planning → Backend Implementation → API Integration → Testing

Key Implementation Areas

- **Auth Architecture:** Secure JWT-based flows with role-specific session handling.
- **Comparison Service:** Field-level matching logic with date tolerance and name normalization.
- **Appeal Workflow:** Notification-driven system for handling verifier disputes and HR responses.
- **Reporting Layer:** Ownership-verified PDF generation and activity statistics.





11. CORA TASK ARCHIVE

- **Authentication:** Handles multi-step user registration, secure OTP generation and distribution via email SMTP, robust JWT session token issuing and validation, and automated session invalidation driven by client inactivity timeouts.
- **Verification:** Executes automated field-level data comparison with date tolerance and name normalization algorithms, dynamically calculates match scores across employee records, and manages persistent verification state in MongoDB.
- **Appeals:** Captures detailed field mismatch logs for disputed records, processes secure supporting document uploads via Vercel Blob storage, and manages end-to-end HR review and approval state transitions.
- **Admin:** Provides full-spectrum HR administration features including interactive KPI dashboard metrics, detailed system access logs with IP tracking, and customized CSV/Excel data export capabilities.
- **Infrastructure:** Establishes core application infrastructure utilizing Mongoose schema design for MongoDB, dedicated email SMTP transport service integration, and Vercel Blob storage for efficient object handling.

12. IMPLEMENTATION MILESTONES

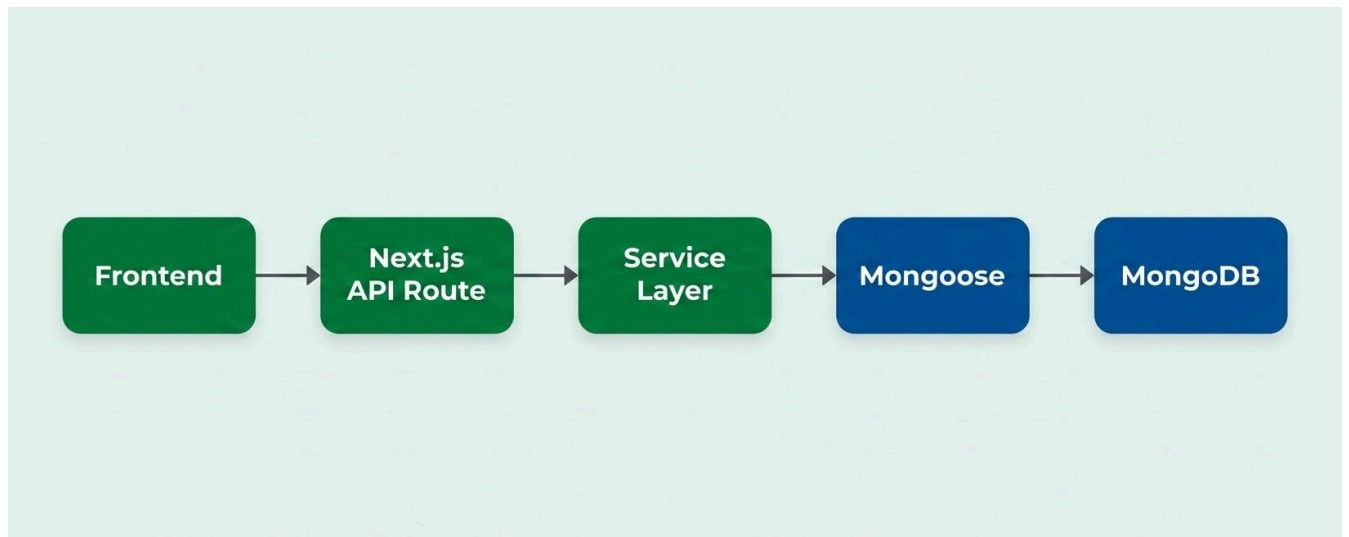
1. **Design:** Established the UI system via Design Mode using a Forest Green (#007A3D) brand theme, Enterprise Blue accents, soft mint backgrounds, and Montserrat/Lato typography with custom DaisyUI components.
2. **Frontend:** Built the interactive verification wizard, public landing page, dual authentication portals, and full HR admin management dashboards using Next.js App Router, React, Tailwind CSS, and Framer Motion.
3. **Authentication:** Secured the platform with JWT session management, bcrypt password hashing, OTP-based two-factor authentication via email SMTP, forgot-password workflows, and automated session inactivity timeouts.
4. **Database:** Modeled all core entities in Mongoose and MongoDB, including schema definitions for Employees, Verifiers, Admins, Verification Records, Appeals, Attempts, Access Logs, and Email Logs.
5. **Verification:** Developed the core automated comparison and scoring engine supporting exact ID matching, case-insensitive name normalization, date tolerance checks, and classification into Matched, Partial Match, or Mismatch results.
6. **Appeals:** Integrated supporting file uploads (via Vercel Blob in production and local storage in development), mismatch collection, verifier comments, and split-panel HR review and approval/rejection response workflows.
7. **Reporting:** Added official PDF report generation with ownership verification via jsPDF/html2canvas and Excel-friendly CSV export capabilities for administrative auditing and metrics.
8. **Email:** Set up Nodemailer over Gmail SMTP for sending 2FA OTPs, welcome messages, automated appeal updates, and report delivery with persistent database logging and metrics tracking.
9. **Audit:** Implemented comprehensive system, access, and action logging that tracks timestamps, user IDs, roles, IP addresses, failure reasons, and user-agent details for compliance.
10. **Security:** Finalized rate limiting on sensitive routes, role-based access control (RBAC), Helmet headers, CORS policies, Joi validation, personal domain rejection during registration, and file size/type validation.

13. BACKEND, DATA & INTEGRATIONS

Core Data Models

- Employee, Verifier, Admin
- Verification Record, Appeal, Verification Attempt
- Access Log, Email Log

Backend Flow



14. AUTHENTICATION & USER MANAGEMENT

The portal relies on JWT authentication and bcrypt hashing, separating verifier and admin access flows to ensure strict data isolation. Tokens include granular scope claims, cryptographic signatures, and configurable expiration windows to protect employee datasets from unauthorized access.

Security Features

- **OTP-Based Two-Factor Authentication (2FA):** Mandatory time-sensitive OTP verification delivered via secure email transport for verifier login workflows.
- **Role-Based Route Protection & JWT Validation:** Strict middleware checking for JWT signatures, issuer parameters, and role claims (Verifier vs. Admin/HR) on all protected endpoints.
- **Session Inactivity Protection:** Automated client-side inactivity timeouts and server-side session invalidation to prevent session hijacking on idle workstations.
- **Rate Limiting on Sensitive Endpoints:** Throttling mechanics applied to login, password reset, and OTP generation requests to protect against brute-force attacks.
- **Credential Hashing & Domain Validation:** Secure password storage via salted bcrypt hashing, paired with strict registration domain checks to reject personal webmail accounts during verifier onboarding.

15. RESPONSIVE UI & POLISH

The interface scales across desktop, tablet, and mobile using Tailwind CSS and DaisyUI.

15.1 Desktop View

Employee Verification

Verifier LoginAdmin Login

Secure & Instant Employee Verification

A fast, reliable, and automated platform for third-party verifiers to confirm employment history. Register your company to get started.

Secure

Fast

Automated

Create a Verifier Account

Company Name *

Your Company Inc.

Your company's official name

Are you a background verification agency? *

Yes

No

Company Email Address *

verifier@yourcompany.com

Only company email domains are allowed (no Gmail, Yahoo, etc.)

Password *

Min. 6 characters

Minimum 6 characters long

Create Account

OR

Already have an account? [Login Now](#)

Made with CodeMate.ai

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15.2 Tablet View

Employee Verification

Verifier LoginAdmin Login

Verifier Login

Two-factor authentication required: Password + OTP

Email Address

your.email@company.com

Password

••••••••

👁

Login

Forgot Password?

Demo Credentials Autofill

EMAIL ADDRESS

sathish.shanmugavel@tvscredit.com

PASSWORD

DemoPassword123!

Use these credentials to explore the app without creating a new account.

New Here?

Don't have an account? [Create one now](#)

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15.3 Mobile View

Emp Verification

Secure & Instant Employee Verification

Fast, automated employment verification for third-party verifiers. Register below to start.

 Secure

 Fast

 Automated

Create a Verifier Account

Company Name *

Your Company Inc.

Your company's official name

BG verification agency? *

☐ Yes

☐ No

Company Email Address *

verifier@yourcompany.com

Company email required (no Gmail/Yahoo)

Password *

Min. 6 characters

Minimum 6 characters long

Create Account

Made with 